

AI-Powered Exam Proctoring & Assessment Engine

Architecture & Technical Stack Specification | Springboard Internship 2026

1. Frontend Tier (UI & Client-Side Logic)

Domain / Component	Technology / Library	Purpose & Implementation
Framework	Next.js 14 (App Router)	High-performance React framework with hybrid SSR/CSR architecture and routing.
Language & Core UI	React 18 + TypeScript	Type-safe, component-driven reactive user interface.
Styling & Theme	Tailwind CSS v3 + PostCSS	CyberHUD dark theme, ambient aurora animations, glassmorphism UI.
Iconography	Lucide React	Icons for controls, timers, dashboard stats, and violation badges.
Code Editor	Monaco Editor (@monaco-editor/react)	VS Code browser IDE engine with multi-language syntax highlighting & indentation.
Client AI Vision	Google MediaPipe (@mediapipe/tasks-vision)	Real-time client-side face landmarking, eye gaze, and head pose estimation.
Internationalization	i18next + react-i18next	Seamless multi-language switching across 6 global locales.
Validation & Auth	Zod + jose + bcryptjs	Schema validation, client-side cryptographic hashing, and JWT handling.
PWA & Offline	Progressive Web App Architecture	Service workers enabling offline resilience and connectivity recovery.

2. AI Proctoring & Computer Vision Pipeline

Domain / Component	Technology / Library	Purpose & Implementation
Object Detection	Ultralytics YOLOv8 (yolov8n.pt)	Sub-15ms real-time detection of phones, tablets, laptops, books & earbuds.
Deep Face Tracking	OpenCV YuNet (ONNX runtime)	Lightweight deep neural network face detector for instant identity verification.
Cascade Face Tracker	OpenCV Haar Cascades	Multi-angle facial landmark verification fallback (frontal + profile XMLs).
Gaze & Orientation	MediaPipe Face Mesh	Calculates eye gaze deviation and suspicious head turning in degrees.
Violation Engine	Custom Proctor Vision (proctor_vision.py)	Automated anomaly flags: Candidate Absence, Multi-Person, Device Detection, Tab Switch.

3. Generative AI & Automated Evaluation Engine

Domain / Component	Technology / Library	Purpose & Implementation
LLM Core Engine	Google Gemini AI (google-genai SDK)	Intelligent question synthesis, subjective grading, and constructive feedback.
PDF Ingestion	pypdf + Gemini Vision/Text	Automated extraction of MCQs, coding prompts, & answer keys from PDF exam sheets.
Subjective Grading	Semantic Evaluator (ai_evaluator.py)	Rubric-grounded grading of descriptive answers with concept coverage scoring.
Translation Service	Custom AI Translator (translator.py)	Real-time multi-language translation of questions, options, and explanations.

4. Multi-Language Sandbox Code Execution Engine

Domain / Component	Technology / Library	Purpose & Implementation
Supported Runtimes	Python 3, JavaScript (Node.js), C, C++, Java	Multi-language compilation and execution runner.
Execution Runner	Subprocess Isolation (code_executor.py)	Sandboxed child process execution per test case with strict resource boundaries.
Timeout Guard	2.5s Strict Execution Limit	Guards against infinite loops, excessive recursion, and Denial-of-Service code.
Verification	Automated I/O Matcher	Standard Input/Output comparison against public and hidden test cases.

5. Backend REST API & Business Logic Layer

Domain / Component	Technology / Library	Purpose & Implementation
API Framework	FastAPI (Python)	High-throughput asynchronous REST API backend.
ASGI Server	Uvicorn (Standard)	Production ASGI server powering fast concurrent request dispatching.
Data Modeling	Pydantic v2	Strict serialization, request validation, and OpenAPI documentation schema.
Security & RBAC	OAuth2 with JWT + Bcrypt	Role-Based Access Control enforcing Student, Examiner, and Admin privileges.
Admin Console	SQLAdmin	Web administrative portal for direct entity monitoring and inspection.

6. Database & Persistent Storage Layer

Domain / Component	Technology / Library	Purpose & Implementation
Database Engine	SQLite (proctoring.db) / PostgreSQL	Relational persistence for users, exams, submissions, logs, and questions.
ORM Layer	SQLAlchemy 2.0	Enterprise Object-Relational Mapper with declarative relational mappings.
Schema Migrations	Alembic	Automated, version-controlled database schema migrations.